

Pricing Feedback and Uniqueness in Combinatorial Entry Games*

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Abstract

Combinatorial entry decisions and endogenous pricing form a feedback loop: which sites a firm activates depends on rivals' prices and markups, and prices and markups depend on the active set. We develop a threshold-map approach for combinatorial entry games with private fixed-cost shocks that turns equilibrium selection into a contraction test. Within a maintained class of site-independent threshold strategies, the equilibrium object is a vector of activation probabilities, and uniqueness follows when the threshold map is a contraction at its fixed points. The map's Jacobian separates private-cost smoothing from marginal-value sensitivity, yielding a *bathtub*-shaped contraction bound in the private-cost noise scale: contraction holds at fixed points in both the large- and small-noise tails, and multiplicity can emerge only at intermediate scales. We verify the required conditions in a Fréchet–CES–Atkeson–Burstein benchmark and quantify the mechanism in global cloud infrastructure entry. In the preferred gravity calibration, replacing endogenous Atkeson–Burstein markups with the constant CES markup eliminates 23% of the active provider-country pairs (from 123 to 95) and reduces aggregate profit by 33.8%. Endogenous pricing therefore sustains marginal deployments on the extensive margin. Limiting markups, typically viewed as a way to address distortions, costs market access.

Keywords: entry games, equilibrium uniqueness, combinatorial discrete choice, Bayesian Nash equilibrium, variable markups

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1. INTRODUCTION

Cloud providers choose deployment portfolios, not isolated markets. A provider’s cloud region in Germany can improve access throughout Europe; one in Singapore can serve Southeast Asia; and one in the United Arab Emirates can function as a hub for nearby markets. The value of each deployment therefore depends on the rest of the provider’s network. It also depends on rivals’ networks: once providers choose their deployment portfolios, the resulting market shares determine prices, markups, and profits. Entry decisions and pricing decisions are therefore jointly determined.

This paper studies combinatorial entry games with endogenous pricing. The central difficulty is a feedback loop. Activation changes the portfolio of sites from which a firm can serve demand, and that portfolio changes market shares and markups through the pricing equilibrium. Variable profit then feeds back into whether other candidate sites remain worth activating. The empirical entry literature (Bresnahan and Reiss, 1991; Berry, 1992; Ciliberto and Tamer, 2009; Seim, 2006; Nishida, 2015) has long confronted the resulting multiplicity, typically by imposing equilibrium-selection rules or bounding parameters via partial identification.

Existing literatures address different parts of the loop. On the private-information entry margin, Espin-Sanchez et al. (2023) prove Bayesian Nash equilibrium uniqueness in binary entry games but cannot capture within-firm complementarities across actions. Doraszelski and Satterthwaite (2010) show in a dynamic industry setting that private payoff shocks generate cutoff strategies without eliminating multiplicity. On the combinatorial portfolio margin, the combinatorial discrete choice literature (Jia, 2008; Arkolakis et al., 2023; Antràs et al., 2017), building on the supermodular-games foundations of Milgrom and Shannon (1994), handles subset selection with interdependent choices but typically works with constant markups. On the endogenous pricing margin, the variable-markup literature (Atkeson and Burstein, 2008; Edmond et al., 2015) endogenizes pricing and market power but conditions on the active set. Our setting combines the three margins, but private information alone does not break the feedback loop. Fixed-cost shocks smooth each firm’s own activation decisions; they do not discipline how strongly rivals’ activations move markups and profits in the pricing equilibrium. A uniqueness argument therefore requires more than private-cost noise. It requires a way to measure whether the markup

and profit responses to rivals' entry amplify or dampen as activation probabilities shift.

We develop a threshold-map approach. Firms observe site-specific private fixed-cost shocks and use site-independent threshold strategies: cutoffs that depend on beliefs about activation probabilities and on the firm's expected portfolio value, but not on the realized shocks at the firm's other sites. A site is activated when its own private-cost draw falls below its cutoff. This maintained restriction turns the game into a continuous fixed-point problem. The equilibrium object is a vector of activation probabilities, and the best-response object is a threshold map from beliefs about activation probabilities into new activation probabilities.

Uniqueness reduces to a contraction test for this threshold map. The map's Jacobian separates three economic forces. The first is private-cost noise: when noise is large, best responses are smooth. The second is marginal-value sensitivity: a perturbation in activation probabilities can move profits by changing market structure and prices. The third is threshold location: sites that are clearly profitable or clearly unprofitable are less responsive than sites close to indifference. The resulting contraction modulus has a bathtub shape in the private-cost noise scale. Large noise smooths best responses directly; small noise delivers contraction only after the relevant region is localized away from marginal sites; and multiplicity can arise only in the middle, where pricing feedback is strong and noise does not yet smooth it out.

The theorem is demand-system agnostic. To make its objects operational, we verify them in a Fréchet–CES–Atkeson–Burstein benchmark with endogenous markups. The benchmark delivers a smooth pricing selection, a bound on the marginal-value Jacobian, and a localized small-noise argument around an isolated complete-information equilibrium.

Endogenous pricing feeds into entry through an amplification term. Near the edge of stability, a small change in a rival's activation triggers outsized price and markup responses; well inside the stable region, those responses die out. This is the endogenous-markup counterpart to the constant-markup combinatorial logic of Antràs et al. (2017): there, activation changes only residual demand; here, it changes both residual demand and the markups used to evaluate marginal sites.

We then quantify the mechanism in global cloud infrastructure entry. The application studies five hyperscale providers—Amazon Web Services, Microsoft Azure, Google

Cloud, Oracle Cloud, and Alibaba Cloud—choosing which of 42 countries to equip with cloud regions. Cloud infrastructure is a natural portfolio-entry environment. Regions affect latency, data residency, compliance, and practical serviceability in nearby markets. A provider’s region in one country can therefore raise the value of other regions in its network or change the firm’s competitive position in markets that are not physically located in that country. This emphasis on asymmetric serviceability is consistent with recent evidence that cloud providers follow a more concentrated location logic than third-party data centers (Fang and Greenstein, 2025).

The data consist of 210 provider-country cells, of which 129 are observed active. We construct a public-data gravity layer that separates market size from provider-specific serviceability. Market size is anchored by GDP and internet penetration. Serviceability is constructed from distance, same-country access, and policy or residency shifters, aggregated through a smooth log-sum accessibility index. Conditional on this gravity layer, the fixed-cost block is calibrated from the observed activation matrix. The preferred gravity calibration matches 200 of 210 provider-country cells.

Pricing feedback is quantitatively first order. Holding fixed the calibrated accessibility and fixed-cost blocks, replacing endogenous Atkeson–Burstein markups with the constant CES markup reduces active provider-country pairs from 123 to 95 and lowers aggregate profit by 33.8%. Endogenous markups therefore sustain 28 additional active provider-country pairs in the calibrated cloud economy. Variable markups raise profits on inframarginal high-share positions and compress dominant market shares relative to the constant-markup benchmark. This profit buffer supports marginal deployments, so when the constant CES markup replaces the endogenous markup rule, profits fall and a set of marginal provider-country pairs exits.

The same deployment margin is relevant for policy counterfactuals. Cloud infrastructure is increasingly shaped by regional rules on data protection, cross-border data transfers, switching, interoperability, and digital sovereignty, especially in Europe (Demirer et al., 2024; Gans et al., 2023; Manganelli and Schnurr, 2024). In an auxiliary EU compliance exercise, a compliance wedge equal to about 11% of the median market-level fixed cost reduces active provider-country pairs from 123 to 110; wedges five and ten times that size reduce them further to 95 and 94. A companion China access exercise, discussed in Section 4.5, treats digital fragmentation as a feasible-access margin rather than only a cost

wedge (Sun and Trefler, 2023; Sun, 2024): in the application, lowering a China-specific cost wedge alone leaves the active set unchanged, while expanding feasible access raises total activation. These exercises show that compliance, residency, and access frictions can reshape market access through the deployment margin.

The paper makes two contributions. First, it provides a threshold-map uniqueness theorem for combinatorial entry games with private information that separates equilibrium selection from the demand system and reduces uniqueness to smoothness, bounded marginal-value sensitivity, and small-noise localization. We verify the theorem’s objects analytically in a Fréchet–CES–Atkeson–Burstein benchmark, showing how endogenous pricing enters the extensive-margin Jacobian through a Perron–Frobenius pricing amplification term. Second, we quantify the pricing-feedback mechanism in global cloud infrastructure entry, where endogenous markups sustain broader deployment than the constant-markup counterfactual and regional compliance frictions can quickly move the extensive margin.

The paper connects two quantitative literatures: relative to combinatorial-entry models that impose equilibrium-selection assumptions (Holmes, 2011; Wollmann, 2018; Fan and Yang, 2020), we supply primitive conditions for uniqueness; relative to multinational-production models that endogenize active locations but keep CES markups (Tintelnot, 2017; Antràs et al., 2026; Castro-Vincenzi, 2024), we close the markup loop.

The theorem establishes uniqueness within the site-independent threshold-strategy class. The benchmark verifies the theorem’s objects analytically in a clean endogenous-markup environment, and the cloud application quantifies the same mechanism in a calibrated setting. This separation keeps distinct what is proved, what is verified, and what is quantified.

The rest of the paper proceeds as follows. Section 2 develops the abstract threshold-map framework. Section 3 verifies the theorem’s objects in the Fréchet–CES–Atkeson–Burstein benchmark. Section 4 presents the cloud infrastructure application and the AB-versus-CES counterfactual. Section 5 concludes.

2. THRESHOLD-UNIQUENESS FRAMEWORK

This section states the equilibrium-selection argument without committing to a particular demand or pricing system. The key object is a threshold map. Given beliefs about which firm-site pairs are active, the map returns the activation probabilities implied by firms' site-level cutoff rules. Equilibrium is a fixed point of this map. Uniqueness follows when the map is a contraction.

The abstraction is useful because the same fixed-point logic can be paired with different pricing environments. Section 3 verifies the required objects in a Fréchet–CES–Atkeson–Burstein benchmark. Section 4 then uses the same threshold logic in the cloud application.

Setup. Consider a combinatorial entry game with F firms. Firm f has a finite set of candidate sites $\mathcal{A}_f$, and

$$M \equiv \sum_{f=1}^F |\mathcal{A}_f|$$

denotes the total number of firm-site pairs. We write a generic firm-site pair as $m = (i, f)$ when this avoids notational clutter.

Firm f privately observes site-specific fixed-cost shocks $\{\varepsilon_{if}\}_{i \in \mathcal{A}_f}$. These shocks are i.i.d. across firms and sites and are drawn from

$$G_{\sigma_\varepsilon}(\varepsilon) = G_0(\varepsilon/\sigma_\varepsilon),$$

where G_0 is a standardized continuously differentiable CDF on $\mathbb{R}$, with density g_0 , and $\sigma_\varepsilon > 0$ is the private-cost noise scale. Activating site i costs $(\phi_i + \varepsilon_{if})w_i$, where ϕ_i and w_i are common knowledge.

We restrict attention to site-independent threshold strategies. Firm f 's strategy is a vector of cutoffs $\{\bar{\varepsilon}_{if}\}_{i \in \mathcal{A}_f}$ such that site i is activated if and only if

$$\varepsilon_{if} \leq \bar{\varepsilon}_{if}.$$

The cutoff for site i may depend on equilibrium beliefs about activation probabilities and on the expected value of the firm's portfolio, but the activation decision for site i does not condition on the realized private-cost shocks of other own sites. Thus the restriction

rules out fully contingent plans over all own-site shock realizations. Throughout the paper, a threshold-strategy BNE means mutual best response within this maintained site-independent threshold-strategy class.¹

The threshold map. Let $v \in [0, 1]^M$ collect activation probabilities for all firm-site pairs. Let

$$v : \mathcal{N} \rightarrow \mathbb{R}^M$$

be a C^1 marginal-value map on an open neighborhood $\mathcal{N}$ of $[0, 1]^M$. Given beliefs v , site m is activated when

$$\varepsilon_m \leq v_m(v).$$

In words, $v_m(v)$ is the largest private-cost shock for which activating site m is still worthwhile under beliefs v ; the firm opens site m whenever its realized shock lies below this cutoff.² The activation probability induced by this cutoff is therefore

$$G_0\left(\frac{v_m(v)}{\sigma_\varepsilon}\right).$$

Define the threshold map

$$T_{\sigma_\varepsilon} : [0, 1]^M \rightarrow [0, 1]^M, \quad T_{\sigma_\varepsilon, m}(v) = G_0\left(\frac{v_m(v)}{\sigma_\varepsilon}\right), \quad m = 1, \dots, M. \quad (2.1)$$

Write $J_T(v)$ for the Jacobian of T_{σ_ε} at v , with σ_ε -dependence suppressed, and $J_v(v)$ for the Jacobian of v . Throughout, $\|x\|_\infty \equiv \max_m |x_m|$ denotes the vector max-norm on $\mathbb{R}^M$; for a matrix A , $\|A\|_\infty$ denotes the induced operator norm, equal to the maximum absolute row sum of A ; and for a bounded function f , $\|f\|_\infty \equiv \sup |f|$ denotes the sup norm. The

¹The restriction rules out strategies in which firm f conditions its decision on site i on the realized own-shocks at other sites—e.g., “activate site i only if ε_{jf} is also low.” Because own-site costs enter additively in the shock and rivals enter payoffs only through activation-probability beliefs v , the single-crossing logic of Athey (2001) suggests that threshold strategies contain best responses within a broader monotone class in supermodular entry games with private values.

²The cutoff is expressed in the same units as ε_m and is already net of the deterministic fixed-cost component. A positive cutoff means activation is profitable except under unusually unfavorable private draws.

associated contraction modulus is

$$\gamma(\sigma_\varepsilon) \equiv \sup_{v \in [0,1]^M} \|J_T(v)\|_\infty. \quad (2.2)$$

Lemma 1 (Threshold fixed-point correspondence). *Under the maintained site-independent threshold-strategy restriction, a profile $v \in [0,1]^M$ is a threshold-strategy BNE if and only if it is a fixed point of T_{σ_ε} .*

Proof. Given a threshold-strategy equilibrium, the activation probability of site m is the probability that $\varepsilon_m \leq v_m(v)$. Hence

$$v_m = G_0(v_m(v)/\sigma_\varepsilon) = T_{\sigma_\varepsilon, m}(v)$$

for every m , so $v = T_{\sigma_\varepsilon}(v)$. Conversely, if $v = T_{\sigma_\varepsilon}(v)$, then the cutoff vector $\bar{\varepsilon}_m = v_m(v)$ induces exactly the activation probabilities encoded by v . The corresponding threshold profile is therefore a mutual best response within the maintained class. $\square$

The lemma is the reason the problem becomes tractable. The original game has combinatorial actions, but the maintained threshold class converts equilibrium selection into a continuous fixed-point problem in activation probabilities.

Theorem 2 (Abstract threshold-strategy uniqueness). *Let G_0 be continuously differentiable and let v be C^1 on an open neighborhood of $[0,1]^M$. Then:*

(i) T_{σ_ε} has at least one fixed point on $[0,1]^M$;

(ii) its Jacobian satisfies

$$J_T(v) = \text{diag} \left(\frac{g_0(v_m(v)/\sigma_\varepsilon)}{\sigma_\varepsilon} \right)_{m=1}^M J_v(v); \quad (2.3)$$

(iii) if $\gamma(\sigma_\varepsilon) < 1$, the fixed point is unique. Hence, by Lemma 1, the threshold-strategy BNE is unique within the maintained site-independent threshold-strategy class.

Proof. Existence follows because T_{σ_ε} is a continuous self-map of the nonempty compact convex set $[0,1]^M$, so Brouwer's theorem applies. The Jacobian factorization follows by differentiating $T_{\sigma_\varepsilon, m}(v) = G_0(v_m(v)/\sigma_\varepsilon)$ coordinate by coordinate. Finally, for any

$v, v' \in [0, 1]^M$, the line segment connecting them lies in $[0, 1]^M$, and the fundamental theorem of calculus gives

$$\|T_{\sigma_\varepsilon}(v) - T_{\sigma_\varepsilon}(v')\|_\infty \leq \gamma(\sigma_\varepsilon)\|v - v'\|_\infty.$$

If $\gamma(\sigma_\varepsilon) < 1$, Banach's fixed-point theorem gives uniqueness. $\square$

Corollary 3 (Bathtub uniqueness in the noise-scale tails). *Under the hypotheses of Lemmas 4 and 5, there exist $\sigma_{\text{low}}, \sigma_{\text{high}} \in (0, \infty)$ with $\sigma_{\text{low}} \leq \sigma_{\text{high}}$ such that the threshold-strategy BNE on $[0, 1]^M$ is unique for every*

$$\sigma_\varepsilon \in (0, \sigma_{\text{low}}) \cup (\sigma_{\text{high}}, \infty).$$

The large-noise tail follows from $\gamma(\sigma_\varepsilon) < 1$ (Lemma 4); the small-noise tail follows from contraction of T_{σ_ε} on the localized set S_{σ_ε} , combined with the fixed-point-containment condition (Lemma 5). Uniqueness on the intermediate range $[\sigma_{\text{low}}, \sigma_{\text{high}}]$ is not delivered by either tail bound and must be checked directly, for example by evaluating $\|J_T(v^\star)\|_\infty$ at the candidate fixed point $v^\star$.

Economic content of the Jacobian. The factorization in (2.3) separates the forces that can generate or rule out multiplicity. The diagonal term,

$$\frac{g_0(v_m(v)/\sigma_\varepsilon)}{\sigma_\varepsilon},$$

measures how many sites are close enough to the entry margin to react to a small change in beliefs. If many sites are nearly indifferent, a small perturbation in activation probabilities can move many entry decisions. The matrix $J_v(v)$ measures how strongly those belief changes move marginal values. This is where pricing feedback enters: activation probabilities change market structure, market structure changes prices and markups, and those price changes feed back into the cutoffs.

The threshold location also matters. A site that is clearly profitable or clearly unprofitable is insensitive to a small perturbation in beliefs. A site at the entry margin is not. Thus uniqueness depends not only on the strength of strategic interaction, but also on where the equilibrium thresholds sit relative to the margin.

Scope of the equilibrium object. The theorem is a uniqueness result within the site-independent threshold-strategy class. It does not rule out equilibria in fully contingent strategies, nor does it derive threshold strategies as the only admissible strategies. The value of the restriction is that it turns a combinatorial entry game into a fixed-point problem whose sensitivity can be measured. This is the equilibrium object used in the benchmark and in the cloud application.

The bathtub modulus. The contraction modulus has a characteristic two-sided shape in σ_ε . When private-cost noise is large, best responses are smooth: the density term in (2.3) is small, so $\gamma(\sigma_\varepsilon)$ falls. When private-cost noise is small, best responses can be sharp, but contraction returns if the relevant region is localized away from marginal sites. The difficult region is the middle, where pricing feedback is strong and noise does not yet smooth it out.

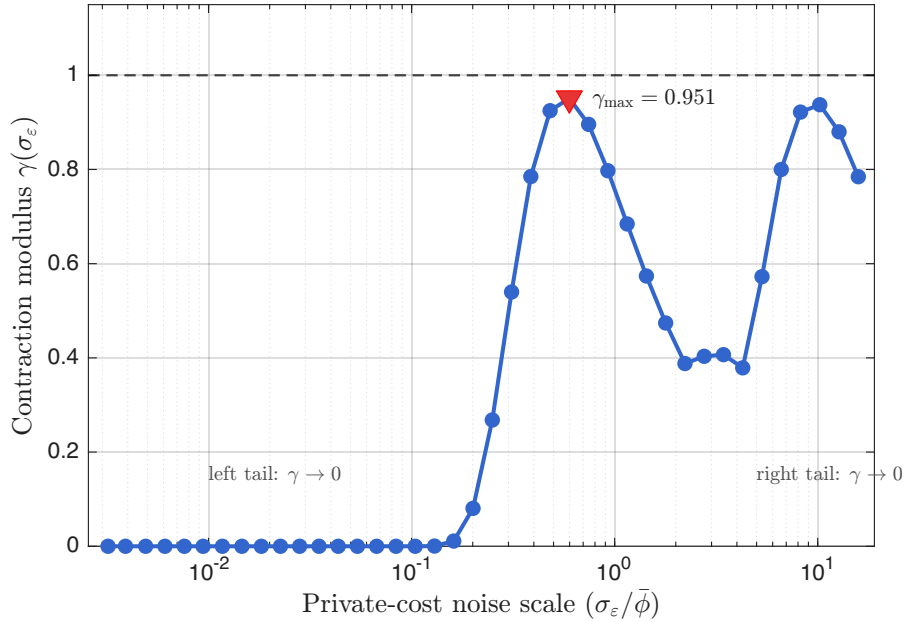


Figure 1: Bathtub-shaped contraction modulus

Operator norm $\|J_T(\nu^*)\|_\infty$ evaluated at the BNE fixed point $\nu^*(\sigma_\varepsilon)$ in a small Fréchet–CES–Atkeson–Burstein verification economy ($N = 3$ markets, $F = 4$ firms; see Section 3). The large-noise tail is disciplined by density dampening; the small-noise tail by localization away from marginal sites. The interior peak $\gamma_{\max} = 0.951$ remains below one across the tested σ_ε grid, so the threshold-strategy BNE in this verification economy is unique at every tested noise scale; uniqueness in the intermediate range is established here by direct evaluation at the fixed point rather than by Corollary 3’s tail bounds.

Figure 1 summarizes this logic. The right tail is a direct density argument. The left tail

is not automatic; it requires a localized region on which threshold indices are uniformly away from zero. The next two lemmas formalize these two tails.

Lemma 4 (Large-noise tail). *Suppose $\|g_0\|_\infty < \infty$ and*

$$K_v \equiv \sup_{v \in [0,1]^M} \|J_v(v)\|_\infty < \infty.$$

Then

$$\gamma(\sigma_\varepsilon) \leq \frac{\|g_0\|_\infty K_v}{\sigma_\varepsilon},$$

so $\gamma(\sigma_\varepsilon) \rightarrow 0$ as $\sigma_\varepsilon \rightarrow \infty$.

Proof. Taking norms in (2.3) gives

$$\|J_T(v)\|_\infty \leq \frac{\|g_0\|_\infty}{\sigma_\varepsilon} \|J_v(v)\|_\infty \leq \frac{\|g_0\|_\infty K_v}{\sigma_\varepsilon}.$$

Taking the supremum over v proves the claim. $\square$

The small-noise tail is more subtle. A positive gap at one fixed point does not by itself make the global modulus small, because the supremum in $\gamma(\sigma_\varepsilon)$ ranges over all off-equilibrium probability profiles. Some off-equilibrium profile may keep a threshold index close to zero. The small-noise argument therefore localizes the contraction check to a set that contains all fixed points and is uniformly non-marginal.

Lemma 5 (Small-noise localization). *Suppose there exists a family $\{S_{\sigma_\varepsilon}\}_{0 < \sigma_\varepsilon < \sigma_0} \subseteq [0,1]^M$ of nonempty, closed, convex sets such that:*

- (i) $T_{\sigma_\varepsilon}(S_{\sigma_\varepsilon}) \subseteq S_{\sigma_\varepsilon}$ for every $\sigma_\varepsilon \in (0, \sigma_0)$;
- (ii) every fixed point of T_{σ_ε} in $[0,1]^M$ lies in S_{σ_ε} ;
- (iii) there exists $b > 0$ such that

$$\inf_{v \in S_{\sigma_\varepsilon}} \min_m |v_m(v)| \geq b$$

for every $\sigma_\varepsilon \in (0, \sigma_0)$;

- (iv) $|z|g_0(z) \rightarrow 0$ as $|z| \rightarrow \infty$;

(v)

$$K_S \equiv \sup_{0 < \sigma_\varepsilon < \sigma_0} \sup_{v \in S_{\sigma_\varepsilon}} \|J_v(v)\|_\infty < \infty.$$

Then

$$\sup_{v \in S_{\sigma_\varepsilon}} \|J_T(v)\|_\infty \rightarrow 0 \quad \text{as } \sigma_\varepsilon \rightarrow 0.$$

In particular, for all sufficiently small σ_ε , T_{σ_ε} is a contraction on $(S_{\sigma_\varepsilon}, \|\cdot\|_\infty)$ and its fixed point is globally unique.

Proof. For $v \in S_{\sigma_\varepsilon}$, condition (iii) implies $|v_m(v)| \geq b$. Hence

$$\frac{g_0(v_m(v)/\sigma_\varepsilon)}{\sigma_\varepsilon} = \frac{1}{|v_m(v)|} \left| \frac{v_m(v)}{\sigma_\varepsilon} \right| g_0\left(\frac{v_m(v)}{\sigma_\varepsilon}\right) \leq \frac{1}{b} \sup_{|z| \geq b/\sigma_\varepsilon} |z| g_0(z).$$

Condition (iv) implies that the right-hand side converges to zero as $\sigma_\varepsilon \rightarrow 0$. Combining this bound with (2.3) and condition (v) gives

$$\sup_{v \in S_{\sigma_\varepsilon}} \|J_T(v)\|_\infty \leq \frac{K_S}{b} \sup_{|z| \geq b/\sigma_\varepsilon} |z| g_0(z) \rightarrow 0.$$

For small enough σ_ε , the map is therefore a contraction on S_{σ_ε} . Condition (i) makes the set invariant, and condition (ii) ensures that every global fixed point lies in that set. Local uniqueness on S_{σ_ε} therefore implies global uniqueness. $\square$

The tail condition $|z|g_0(z) \rightarrow 0$ is weaker than log-concavity. It is satisfied by standard distributions such as the normal and logistic distributions, and by Student- t distributions with degrees of freedom greater than one.

Remark 6 (Verification strategy). The theorem reduces uniqueness to three objects:

- (a) a C^1 marginal-value map v ;
- (b) a bounded marginal-value Jacobian J_v ;
- (c) a localized, invariant, fixed-point-inclusive, uniformly non-marginal region for the small-noise tail.

A unique complete-information equilibrium with a positive marginal-value gap naturally suggests such a localized region, but the gap alone is not enough. One must also verify

that the region is invariant and contains all fixed points. These are environment-specific checks.

Section 3 performs these checks analytically in the Fréchet–CES–Atkeson–Burstein benchmark. Section 4 then uses the same threshold-system logic in the calibrated cloud application, where the pricing and localization objects are assessed through the solved specification and supporting diagnostics.

3. BENCHMARK VERIFICATION: FRÉCHET–CES–AB

Assumption 7 (Private cost shocks and threshold strategies). Fixed-cost shocks $\{\varepsilon_{if}\}$ are i.i.d. across firm-site pairs, with standardized CDF G_0 , bounded density $g_0 = G'_0$, scale $\sigma_\varepsilon > 0$, and $|z|g_0(z) \rightarrow 0$ as $|z| \rightarrow \infty$ when invoking the small-noise tail. Firm f observes its own shocks but not rivals' shocks. Firm f uses the site-independent threshold strategies defined in Section 2.

The abstract theorem leaves the marginal-value map v unspecified. This section verifies its required objects in a Fréchet–CES–Atkeson–Burstein benchmark with endogenous markups. The benchmark's role is to provide a clean environment in which smoothness, bounded marginal-value sensitivity, and small-noise localization can be checked analytically.

The verification has three parts. The Fréchet cost aggregator and the two-tier CES demand system deliver a tractable marginal-value map. Cournot pricing delivers the Atkeson–Burstein markup rule and, under a Perron–Frobenius pricing contraction, a unique smooth pricing selection. An isolated complete-information equilibrium then anchors the localized small-noise argument. Together, these steps verify the three ingredients from Section 2.

3.1 Benchmark environment

There are F firms, indexed by f , and N markets, indexed by n . Firm f has a finite set of candidate sites $\mathcal{A}_f$. An activation set is denoted $\mathcal{I}_f \subseteq \mathcal{A}_f$. In each market, final demand follows a two-tier CES aggregator with outer elasticity $\eta > 1$ and inner, within-sector elasticity $\sigma > \eta$. Sectoral expenditure R_n is taken as exogenous.

Firm f 's marginal cost of serving market n from activation set $\mathcal{I}_f$ is

$$c_{nf}(\mathcal{I}_f) = \frac{\Gamma_\theta}{a_f} \Sigma_{fn}(\mathcal{I}_f)^{-1/\theta}, \quad \Sigma_{fn}(\mathcal{I}_f) = \sum_{i \in \mathcal{I}_f} (w_i \tau_{in})^{-\theta}. \quad (3.1)$$

Here a_f is firm productivity, w_i is the input cost at site i , τ_{in} is the trade or access cost from site i to market n , and θ is the Fréchet dispersion parameter. We maintain $(\sigma - 1)/\theta > 1$, so that the Fréchet dispersion is small relative to the within-sector demand elasticity. If $\mathcal{I}_f = \emptyset$, we set $c_{nf} = +\infty$ and $VP_{nf} = 0$, so firm f is absent from the pricing block in market n .

Throughout this section, σ denotes the within-sector CES elasticity. The private fixed-cost noise scale remains σ_ε .

Given active firms, firms compete à la Cournot in each market, yielding the Atkeson–Burstein markup rule

$$p_{nf} = \mu(s_{nf}) c_{nf}, \quad \mu(s) = \frac{\epsilon(s)}{\epsilon(s) - 1}, \quad \frac{1}{\epsilon(s)} = \frac{1 - s}{\sigma} + \frac{s}{\eta}, \quad (3.2)$$

where s_{nf} is firm f 's revenue share in market n . Markups are bounded:

$$\mu(s) \in \left[\frac{\sigma}{\sigma - 1}, \frac{\eta}{\eta - 1} \right].$$

Thus high-share firms face a lower perceived elasticity and charge a higher markup, but the markup schedule remains uniformly bounded.

Firm f 's ex-post profit, conditional on an activation profile and its private cost draws, is

$$\pi_f(\mathcal{I}_f, \mathcal{I}_{-f}; \varepsilon_f) = \sum_n VP_{nf}(\mathcal{I}_f, \mathcal{I}_{-f}) - \sum_{i \in \mathcal{I}_f} (\phi_i + \varepsilon_{if}) w_i,$$

where VP_{nf} is variable profit in market n . The private fixed-cost shocks ε_{if} are observed by firm f before entry decisions are made.

The benchmark contains the pricing feedback absent from constant-markup combinatorial discrete-choice problem (CDCP) models. Activating a site changes the firm's cost of serving markets. Costs change market shares. Market shares change AB markups. Those markups then feed back into the value of activating other sites. The rest of the section

verifies when this loop is disciplined.

3.2 The pricing block

Fix an activation profile. The pricing equilibrium can be written as a finite fixed point in rival-demand indices. Let $D_{-f}(n)$ denote the rival demand index faced by firm f in market n . For a pricing state D , define the pointwise interaction matrix

$$\Lambda(D)_{(n,f),(n,g)} = \kappa_{nf}(D) \frac{p_{ng}(D)^{1-\sigma}}{D_{-f}(n; D)}, \quad g \neq f, \quad (3.3)$$

with diagonal and cross-market entries equal to zero. Here

$$\kappa_{nf}(D) = \left| \frac{\partial p_{nf}(D)^{1-\sigma}}{\partial D_{-f}(n)} \right|$$

is the pricing-feedback coefficient. If a market has only one active firm, the corresponding row is zero because there is no rival-demand interaction.

The matrix $\Lambda(D)$ measures how a perturbation to one firm's pricing problem propagates through rivals. If the perturbation dies out under iteration, the pricing block is stable. If it is amplified, the pricing block is close to unstable and the entry system inherits that sensitivity.

The benchmark uses a uniform domination condition. Let Ψ be the pricing update map in rival-demand indices. Suppose there exists a nonnegative matrix $\bar{\Lambda}$ such that

$$|J_{\Psi}(D)| \leq \bar{\Lambda}$$

throughout the relevant compact pricing domain. The Perron–Frobenius spectral radius of this envelope is the pricing-contraction object.

Proposition 8 (Pricing uniqueness). *Fix an activation profile. Suppose there exists a nonnegative matrix $\bar{\Lambda}$ such that $|J_{\Psi}(D)| \leq \bar{\Lambda}$ for every pricing state D in the relevant compact pricing domain. If*

$$\rho(\bar{\Lambda}) < 1,$$

then the AB pricing equilibrium is unique. Moreover, the pricing selection is smooth in the cost

vector on the maintained interior region. A direct sup-norm sufficient condition is

$$\sup_D \|J_\Psi(D)\|_\infty < 1;$$

under the normalization in (3.3), this follows from $\sup_D \max_{n,f} \kappa_{nf}(D) < 1$.

The proof is in Appendix A.1. The key step is a Perron–Frobenius weighted sup-norm contraction. If $\rho(\bar{\Lambda}) < 1$, there exists a weighted norm in which the pricing update map is a contraction. Banach’s theorem gives a unique pricing fixed point, and the implicit-function argument gives smooth dependence of equilibrium prices on costs.

Proposition 8 has two roles. On the intensive margin, it gives the smooth pricing selection needed to define a C^1 threshold map. On the extensive margin, the same contraction object controls how strongly activation probabilities can move prices and markups. For any $\lambda_\Lambda \in (\rho(\bar{\Lambda}), 1)$, pricing responses are bounded by a term proportional to

$$\frac{1}{1 - \lambda_\Lambda}.$$

This is the pricing-feedback amplification term. When the pricing block is far from the boundary, the amplification is small. When the pricing block is close to unstable, even small activation perturbations can generate large price and markup responses.

This is the benchmark’s main departure from the constant-markup CDCP logic in Antràs et al. (2017). With constant markups, an activation perturbation changes residual demand but not the markup schedule. With AB markups, the same perturbation also changes the elasticity each firm faces. The extensive margin therefore inherits the pricing feedback summarized by $\bar{\Lambda}$.

3.3 The benchmark threshold map

We now construct the benchmark counterpart of the abstract marginal-value map v . Fix beliefs $\nu \in [0, 1]^M$ over firm-site activation probabilities. Under the site-independent threshold restriction, all firm-site activations other than the candidate site (i, f) are independent Bernoulli draws with probabilities given by ν .

Let $\mathcal{I}_f^{-i}$ denote firm f ’s activation set excluding site i . Define the deterministic profit

increment from activating site i , net of the candidate site's private shock, by

$$\Delta\Pi_{if}^0(\mathcal{I}_f^{-i}, \mathcal{I}_{-f}) = \sum_n \left[VP_{nf}(\mathcal{I}_f^{-i} \cup \{i\}, \mathcal{I}_{-f}) - VP_{nf}(\mathcal{I}_f^{-i}, \mathcal{I}_{-f}) \right] - \phi_i w_i.$$

The superscript 0 indicates that the deterministic fixed cost is included, while the candidate site's private-shock cost $\varepsilon_{if} w_i$ is excluded. Site i is profitable for firm f when

$$\varepsilon_{if} \leq \frac{\Delta\Pi_{if}^0}{w_i}.$$

The benchmark marginal-value index is therefore

$$v_{if}(v) = \frac{\mathbb{E}_v \left[\Delta\Pi_{if}^0(\mathcal{I}_f^{-i}, \mathcal{I}_{-f}) \right]}{w_i}, \quad (3.4)$$

where the expectation is over the Bernoulli activation profile induced by v and over the unique AB pricing equilibrium for each realized active set. The threshold map is exactly the abstract map from Section 2:

$$T_{\sigma_\varepsilon, if}(v) = G_0 \left(\frac{v_{if}(v)}{\sigma_\varepsilon} \right).$$

Equation (3.4) is where the benchmark meets the theorem. The map v is a literal expectation over realized activation profiles. The pricing equilibrium is solved inside each realized profile, and the Bernoulli probabilities enter as weights. Once the pricing selection is unique and smooth, the same is true for v .

Let $x^{CI} \in \{0, 1\}^M$ denote the complete-information activation vector when private shocks vanish. If the complete-information equilibrium is unique and all site-level marginal values are strictly separated from zero, define the complete-information marginal-value gap by

$$\delta_v = \min_m |v_m(x^{CI})|.$$

A positive gap gives local sign stability, but it does not by itself rule out fixed points far from x^{CI} . The small-noise tail therefore needs one additional exclusion condition.

Assumption 9 (Benchmark isolation). There exist $r \in (0, 1/2)$ and $b > 0$ such that, outside

the cube

$$S = \{v : \|v - x^{CI}\|_\infty \leq r\},$$

at least one coordinate $m = m(v)$ satisfies

$$|v_m(v)| \geq b, \quad |\mathbf{1}\{v_m(v) > 0\} - v_m| \geq 2r.$$

The condition says that any profile outside the local neighborhood of the complete-information equilibrium makes a clear mistake in at least one coordinate: the coordinate is non-marginal, and the probability assigned to it is separated from the sharp complete-information best response. This prevents off-equilibrium probability profiles from generating additional small-noise fixed points.

3.4 Verification theorem

The next result collects the benchmark verification. It maps the objects in the Fréchet–CES–AB environment into the three requirements of Theorem 2, Corollary 3, and Lemmas 4–5.

Proposition 10 (Benchmark verification of the threshold-map conditions). *In the Fréchet–CES–Atkeson–Burstein benchmark, suppose Assumption 7 holds and the pricing block satisfies the uniform domination condition in Proposition 8 with $\rho(\bar{\Lambda}) < 1$. Then:*

- (i) *The marginal-value map $v(v)$ in (3.4) is C^1 on an open neighborhood of $[0, 1]^M$.*
- (ii) *For any weighted-norm pricing contraction constant $\lambda_\Lambda \in (\rho(\bar{\Lambda}), 1)$,*

$$K_v \equiv \sup_{v \in [0, 1]^M} \|J_v(v)\|_\infty \leq \frac{L_D L_\mu}{1 - \lambda_\Lambda} + L_C,$$

where L_D , L_μ , and L_C depend only on model primitives (σ, η, θ) , markup bounds, market-size bounds, and the Fréchet cost aggregator.

- (iii) *Suppose the complete-information game has a unique pure-strategy equilibrium x^{CI} with positive marginal-value gap $\delta_v > 0$, and suppose Assumption 9 holds. Then, for all sufficiently small σ_ε , the set S in Assumption 9 is invariant, fixed-point-inclusive, and uniformly non-marginal for the threshold map.*

Consequently, the benchmark verifies the smoothness, bounded-Jacobian, and localized small-noise ingredients of the abstract theorem. If the resulting contraction modulus satisfies $\gamma(\sigma_\varepsilon) < 1$, the threshold-strategy BNE is unique within the maintained site-independent threshold-strategy class. If $\gamma_{\max} \equiv \sup_{\sigma_\varepsilon > 0} \gamma(\sigma_\varepsilon) < 1$, uniqueness holds uniformly over private-cost noise scales.

Proof sketch and interpretation. Part (i) follows because the expectation in (3.4) is a finite Bernoulli mixture over activation profiles, and each realized profile has a unique smooth pricing selection by Proposition 8. Part (ii) separates the derivative of the marginal-value map into a direct cost-aggregation component and an indirect pricing component. The indirect component is bounded by the Perron–Frobenius pricing response, which gives the amplification term $1/(1 - \lambda_\Lambda)$. Part (iii) combines local sign stability around the isolated complete-information equilibrium with the exclusion condition in Assumption 9. Local sign stability keeps the threshold map inside the non-marginal cube for small σ_ε , while the exclusion condition rules out fixed points outside the cube. The full proof is in Appendix A.

The bound in part (ii) is the main economic object. The constants L_D and L_C summarize the direct effect of activation on costs and variable profits. The term $L_\mu/(1 - \lambda_\Lambda)$ summarizes the endogenous-markup channel. When pricing is stable, activation perturbations have limited effects on markups and the threshold map remains insensitive. When pricing is close to the boundary, markup responses amplify the extensive-margin Jacobian. Part (iii) is narrower: a positive complete-information gap gives a natural candidate region, but fixed-point inclusion and invariance require the benchmark isolation condition. The left tail of the bathtub modulus is therefore a localization result rather than an automatic consequence of private-cost noise.

3.5 Portability beyond CES–AB

The verification architecture is not tied to the CES–AB benchmark. The abstract theorem requires a smooth pricing selection, a bound on how equilibrium prices respond to activation perturbations, and a localization argument for the small-noise tail. Online Appendix [OA.2](#) records these pricing-side ingredients for single-product Bertrand pricing under nested-logit demand.

In that environment, the first-order condition can be written as

$$p_{nf} - c_{nf} = \frac{1}{\gamma_n \left[\lambda_{n,k(f)}^{-1} (1 - s_{nf|k(f)}) + s_{nf|k(f)} - s_{nf} \right]}.$$

The aggregative-game argument of Nocke and Schutz (2018) delivers a unique pricing equilibrium under the maintained regularity conditions. Smooth dependence of equilibrium prices on costs then gives a C^1 marginal-value map, while bounded share derivatives and bounded markups give a finite Jacobian bound.

Thus the transferable object is not the CES geography. It is the verification architecture: identify a smooth pricing selection, bound the pricing response to activation perturbations, and combine that bound with a localized non-marginal region for the threshold map. The cloud application keeps this threshold-system logic but replaces the Fréchet cost aggregator with a public-data gravity-accessibility layer.

4. CLOUD INFRASTRUCTURE APPLICATION

This section quantifies the pricing-feedback mechanism in global cloud infrastructure entry. The application studies five hyperscale providers—Amazon Web Services, Microsoft Azure, Google Cloud, Oracle Cloud, and Alibaba Cloud—choosing which countries to equip with cloud regions. The setting is useful for two reasons. First, deployment is a portfolio decision: a region affects latency, serviceability, compliance, and the value of other regions in the same network. Second, pricing is endogenous: a provider’s deployment portfolio changes market shares, market shares change markups, and markups feed back into the profitability of marginal deployments.

The application fixes a public-data gravity-accessibility layer, calibrates deployment frictions from the observed activation matrix, and asks how the active set changes when the pricing rule is changed. The main counterfactual replaces endogenous Atkeson–Burstein markups with the constant CES markup while holding the accessibility and fixed-cost blocks fixed.

4.1 Data and sample

The empirical panel is provider-country. A cell is coded active when a provider operates at least one cloud region in that country. The underlying deployment objects are provider regions, each of which contains one or more availability zones where compute, storage, and networking services are offered from local infrastructure. We compile region lists from each provider’s official documentation (Q1 2026 vintage) and use Datacenter Map (purchased database, April 9, 2026 vintage) as a cross-check and for harmonized country coding. When the two sources disagree, provider documentation takes precedence.

Table 1 summarizes the sample. The panel contains five firms, 42 countries, and 210 provider-country cells. Of these cells, 129 are operationally active in the data. We also record five announced pipeline cells. Pipeline cells enter the maintained feasible set but are coded inactive in the target activation matrix until the deployment is operational.

Table 1: Data and Sample Construction

Firm	Provider	Operational country-cells	Pipeline country-cells	Feasible country-cells
AWS	Amazon Web Services	25	5	30
MSFT	Microsoft Azure	34	0	34
GOOG	Google Cloud	29	0	29
ORCL	Oracle Cloud	27	0	27
BABA	Alibaba Cloud	14	0	14
Total		129	5	134

The maintained candidate sets are lower bounds. They include operational and pipeline countries but do not add hypothetical locations to improve fit. Thus 134 cells lie inside the feasible-set lower bound, while the remaining 76 cells are outside the maintained candidate sets and are mechanically inactive in the solved model. We therefore report both full-panel fit and feasible-cell fit, with the latter isolating the economically informative part of the classification exercise.

The five-provider sample is also deliberate. AWS, Azure, and Google Cloud account for the dominant share of global cloud infrastructure revenue, but a Big-3 sample would mute the extensive-margin heterogeneity that makes the application useful. Oracle and Alibaba contribute precisely that variation. Alibaba’s footprint is concentrated in East and Southeast Asia, while Oracle operates a broader multi-region rollout into emerging

markets. The resulting feasible-set heterogeneity, with $|\mathcal{A}_f|$ ranging from 14 for Alibaba to 34 for Microsoft, makes the pricing-feedback channel visible across firms rather than only in the aggregate.

4.2 Gravity calibration

Cloud serviceability is not well described by a textbook iceberg geography block. A provider may serve nearby users with low latency, may satisfy data-residency requirements only from a local or regional deployment, and may use one region as a hub for several nearby markets. This emphasis on asymmetric serviceability is consistent with recent evidence that cloud providers follow a more concentrated location logic than third-party data centers (Fang and Greenstein, 2025). We therefore build a public-data gravity layer that separates market size from provider-specific serviceability.

Market demand is captured by a provider-invariant public-data proxy

$$\Omega_n \propto (\text{GDP}_n \times \text{internet penetration}_n)^\beta,$$

normalized to the United States. Provider-market serviceability is constructed from candidate sites. For candidate site i and market n , define the gravity score

$$q_{in} = -\beta_d \log(1 + d_{in}) + \beta_{\text{same}} \mathbf{1}\{i = n\} + \beta_{\text{local}} z_{in}, \quad (4.1)$$

where d_{in} is a latency proxy and z_{in} is an OECD-based policy or residency shifter. Provider f 's accessibility index aggregates active sites through a smooth log-sum,

$$A_{fn}(\mathcal{I}_f) = \lambda_A^{-1} \log \left(1 + \sum_{i \in \mathcal{I}_f} \exp(\lambda_A q_{in}) \right). \quad (4.2)$$

We normalize accessibility against a fixed market-level frontier computed over the common candidate universe:

$$\psi_{fn}(\mathcal{I}_f) = \exp \left(\chi [A_{fn}(\mathcal{I}_f) - A_n^{\text{frontier}}] \right) \leq 1. \quad (4.3)$$

The object ψ_{fn} is a provider-market quality shifter. It captures how well provider f can

serve market n from its deployment portfolio.

The fixed-cost block is

$$FC_{fi} = \phi_i g_f w_i,$$

where ϕ_i is a market-level deployment cost, g_f is a provider-specific fixed-cost shifter, and w_i is the site cost scale. Gravity-layer parameters are chosen on a small plausibility grid rather than estimated from entry moments. The fixed-cost block (ϕ_i, g_f) is then calibrated from the observed activation matrix.

Table 2 summarizes the sources of discipline. Market size and serviceability are anchored outside the binary entry moments. Deployment frictions are calibrated from entry. Holdout and pricing diagnostics are used for validation, not for separate identification.

The gravity layer changes the source of marginal-value variation, not the equilibrium object. The maintained threshold system remains the same as in the theory. What changes is that the application uses a smooth certainty-equivalent accessibility index rather than the literal Fréchet cost aggregator from the benchmark. The implementation should be read as a smooth reduced-form accessibility layer paired with the maintained threshold system, not as a new theorem environment. Online Appendix [OA.3](#) records the corresponding accessibility smoothness, quality-adjusted pricing regularity, full-Jacobian decomposition, and solved-specification diagnostics. Appendix B reports the compact cloud application supplement on sample construction, candidate sets, and preferred calibration. Online Appendix [OA.4](#) reports the remaining extended validation exercises and comparator evidence.

Table 2: Sources of Discipline in the Cloud Application

Object	Economic role	How anchored
Ω_n	market-size scale	public-data proxy based on GDP and internet penetration, normalized externally
q_{in} and ψ_{fn}	geography, latency, and residency access	public geography and policy shifters combined through a smooth gravity-accessibility layer
(ϕ_i, g_f)	deployment-friction block	calibrated from the observed activation matrix conditional on the accessibility layer
Holdout and diagnostics	validation and regularity checks	masked-cell holdout, spectral-radius diagnostics, and Jia–Tarski gap checks

Notes: The table separates externally anchored objects, calibrated objects, and validation objects. The application is a calibrated deployment model designed to isolate the pricing-feedback mechanism.

4.3 Fit, validation, and diagnostics

The preferred gravity baseline uses $\chi = 0.40$, frontier-difference normalization, and the OECD-based “suggested_weight” policy mode. Conditional on those accessibility inputs, recalibrating (ϕ_i, g_f) matches the observed activation matrix well. The model correctly classifies 200 of 210 provider-country cells, or 124 of 134 feasible cells once mechanically inactive outside-feasible cells are set aside. Table 3 reports the main solved-specification diagnostics.

Table 3: Preferred Gravity Baseline: Fit and Diagnostics

Object	Value	Interpretation
Full-panel agreement	200/210	all provider-country cells
Feasible-cell agreement	124/134	excludes mechanically inactive outside-feasible cells
Precision / recall	98.4% / 93.8%	active-cell classification discipline
Holdout agreement	0.819 vs. 0.614	five-fold mean vs. majority-class baseline
$\rho(\Lambda)$	0.1106	solved-specification pricing diagnostic
$\max_{n,f} \kappa_{nf}$	0.2355	pointwise markup-feedback diagnostic
Jia–Tarski gap	0	supremum and infimum solves coincide

Notes: Pricing and Jia–Tarski statistics are evaluated at the solved gravity specification. The holdout exercise is conditional on the maintained feasible-set design and equilibrium structure; fold-level results are reported in Online Appendix [OA.4](#).

Figure 2 displays the observed and fitted activation matrices. The mismatches are concentrated in a small set of regional hub markets, especially the UAE, Saudi Arabia,

Singapore, and Thailand. This is a useful failure mode: the public-data gravity layer captures broad serviceability well, but it does not fully encode the strategic value of a few hub-like locations where network, peering, or regulatory considerations are especially important.

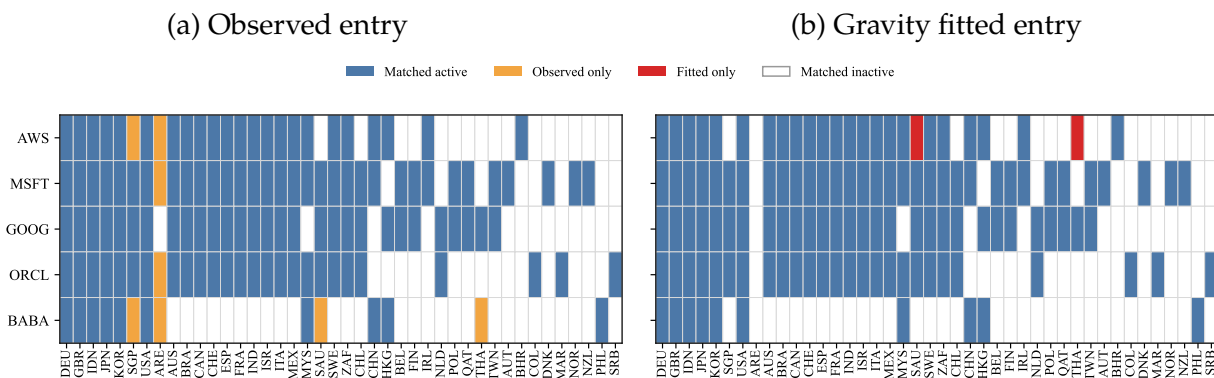


Figure 2: Observed versus fitted activation matrix under the preferred gravity baseline
Blue cells are matched active markets, white cells are matched inactive markets, orange cells are observed-only activations, and red cells are fitted-only activations. The model matches 200 of 210 provider-country cells.

Validation beyond in-sample fit. The in-sample fit is informative but not an independent test of the mechanism, because the fixed-cost block is calibrated to entry moments. We therefore use a five-fold masked-cell holdout exercise to assess predictive discipline within the same design. In each fold, 42 of the 210 cells are held out, (ϕ_i, g_f) is recalibrated on the remaining 168 cells, and the model is solved on the full lattice before the held-out cells are scored. Mean holdout agreement is 0.819, compared with a majority-class baseline of 0.614; the mean holdout true-positive and true-negative rates are 0.747 and 0.936. Online Appendix [OA.4](#) reports the fold-level results and additional validation exercises.

What the holdout exercise does and does not measure. The protocol masks the moment-matching step but not the parameter space. Because ϕ_i is a per-market fixed cost, any unmasked active cell in market i contributes information that pins down ϕ_i during recalibration, and the held-out cell in that same market is then scored against a ϕ_i that the unmasked cells have effectively identified. The reported 0.819 agreement therefore measures the model’s ability to predict held-out activations *conditional on the calibration recovering market-level fixed costs from the remaining cells*, not its ability to predict activations in markets the calibration has not seen. We retain the exercise as a discipline against gross overfitting—a more flexible parameterization could not reach 0.819 even with this leakage—but the number is not interpretable as out-of-sample predictive accuracy in the strong sense.

Table 4: Five-Fold Holdout Validation

Fold	n_{train}	n_{hold}	Train agr.	Holdout agr.	Holdout TPR / TNR
1	168	42	0.940	0.786	0.679 / 1.000
2	168	42	0.946	0.762	0.724 / 0.846
3	168	42	0.952	0.833	0.739 / 0.947
4	168	42	0.952	0.881	0.875 / 0.889
5	168	42	0.946	0.833	0.720 / 1.000
Mean	—	—	0.948	0.819	0.747 / 0.936
SD	—	—	0.005	0.046	— / —
Majority-class baseline				0.614	—

Notes: Each fold masks 42 of the 210 provider-country cells. The model is recalibrated on the remaining 168 cells and then solved on the full lattice before the held-out cells are scored. The majority-class benchmark is 0.614.

Pricing diagnostics remain well behaved across the counterfactuals we study. The

solved-specification spectral radius ranges from 0.095 to 0.122, far below one, across the baseline and main counterfactuals. Online Appendix [OA.3](#) reports the full diagnostic figure and the calibrated-cost pricing audit.

4.4 Endogenous markups sustain broader deployment

Pricing feedback is quantitatively first order. The main counterfactual holds fixed the calibrated accessibility layer and fixed-cost block, and replaces endogenous Atkeson–Burstein markups with the constant CES markup $\mu = \sigma/(\sigma - 1)$.

The preferred gravity baseline has 123 active provider-country pairs. Under the constant-markup CES pricing rule, the equilibrium has 95 active pairs. Replacing the markup rule therefore eliminates 28 provider-country pairs, a 23% decline in the extensive margin, and lowers aggregate profit by 33.8%.

The lost provider-country pairs are best understood as marginal deployments rather than as the core of the cloud network. The AB economy keeps these cells active because endogenous markups generate rents on inframarginal positions and because the AB pricing rule compresses dominant shares relative to the constant-markup economy. Both forces raise the variable-profit buffer available to support sites close to the entry cutoff. When the CES markup replaces the AB rule, this buffer falls. Cells that were active because the portfolio as a whole made them just profitable drop below the threshold.

Figure 3 shows the mechanism. AB markups exceed the constant CES benchmark for active provider-country pairs. Because high-share firms face lower perceived elasticities, the AB rule raises their markups; the same endogenous markup schedule also compresses dominant shares relative to the constant-markup economy. This combination raises profits on inframarginal positions and keeps marginal deployments viable. When the CES markup replaces the AB rule, profits fall for all five providers and a set of marginal deployments exits.

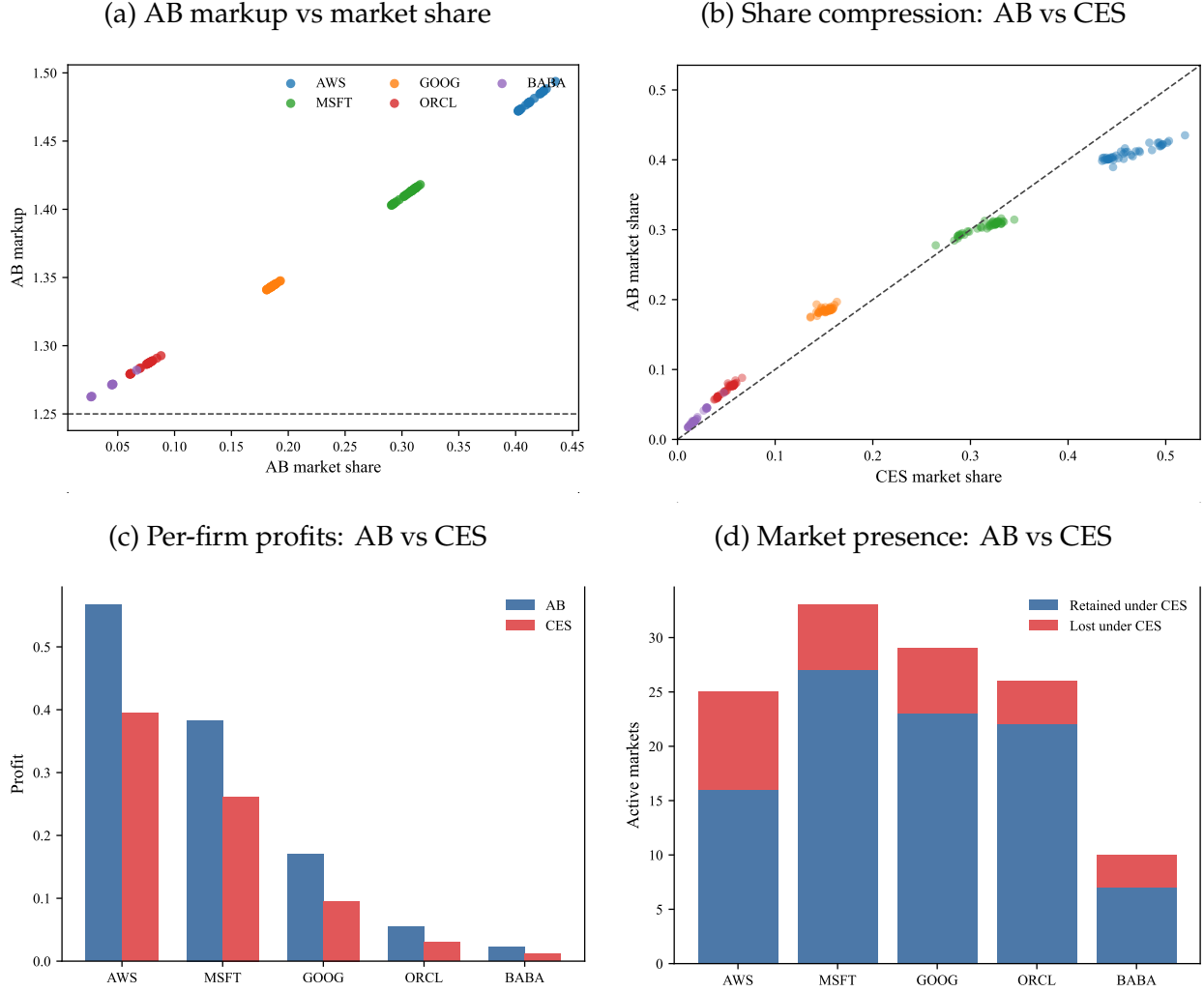


Figure 3: AB versus CES pricing under the preferred gravity baseline

Panel (a) plots AB markups against market shares for active provider-country pairs, with the constant CES markup shown by the dashed line. Panel (b) compares market shares under AB and CES pricing. Panel (c) reports per-firm profits under both pricing rules. Panel (d) decomposes AB-active markets into CES-retained and CES-lost markets by firm.

The per-firm effects are not proportional to firm size. AWS loses the largest number of active cells in the CES counterfactual, but the largest proportional profit declines occur for Google, Oracle, and Alibaba. The profit changes are AWS -30.5% , MSFT -31.7% , GOOG -43.6% , ORCL -46.2% , and BABA -48.0% . This heterogeneity reflects the interaction between markup rents and calibrated deployment costs. Firms with broad portfolios lose markup rents across many cells, while the fixed-cost block determines which marginal cells remain viable after the pricing rule changes.

The result is therefore extensive-margin rather than purely intensive: endogenous

markups sustain deployments whose variable profits are close to the fixed-cost threshold.

Interpretation: mechanism counterfactual, not identified treatment. The exercise compares two pricing regimes within a single calibration: (ϕ_i, g_f) are recovered under the AB pricing rule and held fixed when the constant CES markup is imposed. The reported magnitudes therefore identify the contribution of the markup rule to the active set and to aggregate profits *conditional on the AB-calibrated cost block*, not the deployment that would obtain in an economy whose fixed-cost block was separately calibrated under CES. A symmetric exercise that recalibrates (ϕ_i, g_f) under CES would by construction recover the observed activation pattern up to the model’s fit and is therefore uninformative about the role of pricing feedback. The asymmetry is the point: we ask how much of the deployment pattern in the AB-calibrated economy is sustained by the markup rule itself, holding the rest of the calibration fixed. The 23% and 33.8% magnitudes should be read as the size of that channel under the preferred calibration, not as the gap between two independently identified equilibria.

4.5 Regional compliance and access counterfactuals

The same deployment margin is relevant for policy counterfactuals. We first consider an EU compliance counterfactual. Starting from the preferred gravity calibration, we add an additive compliance wedge $\Delta\phi$ to the market-level fixed-cost component of EU markets while holding fixed the gravity accessibility layer and the calibrated deployment-friction block. We then report the ladder under the preferred AB pricing rule and under the constant-markup CES pricing rule.

The counterfactual is motivated by the way cloud deployment is discussed in current European digital policy. Cloud providers face not only standard competition-policy scrutiny, but also rules and procurement criteria related to data protection, cross-border data transfers, switching, interoperability, and digital sovereignty (Demirer et al., 2024; Gans et al., 2023; Manganelli and Schnurr, 2024). The compliance wedge is a reduced-form way to isolate how this class of regional frictions moves the deployment margin.

Table 5 reports the response in two panels. Panel A summarizes the aggregate response under AB and CES pricing. Under AB, active provider-country pairs fall from 123 at baseline to 110 at $\Delta\phi = 0.001$, a wedge equal to about 11% of the median calibrated

market-level fixed cost, and to 95 and 94 at $\Delta\phi = 0.005$ and $\Delta\phi = 0.010$. Under CES, the same ladder starts from 95 active provider-country pairs and falls to 80 at $\Delta\phi = 0.001$ and to 73 at $\Delta\phi = 0.005$ and $\Delta\phi = 0.010$. In both pricing regimes, the response is steep at small wedges and then saturates quickly.

The steep-then-saturating pattern is informative about the distribution of profit gaps. A small compliance wedge first removes provider-country pairs whose variable profits are already close to the fixed-cost threshold. Once this first wave of marginal EU deployments exits, the remaining EU cells are more inframarginal: they either serve larger markets, have stronger access value, or are supported by a broader provider portfolio. Further increases in the wedge therefore have a much smaller extensive-margin effect. Aggregate producer profit rises even as entry falls, so the EU exercise is best read as an extensive-margin market-access result rather than as a uniform profit-loss story.

Table 5: EU Compliance Counterfactual: Aggregate Response and Incidence

<i>Panel A. Aggregate response</i>							
$\Delta\phi$	Rel./med. ϕ_i	Active		Δn		Δ profit (%)	
		AB	CES	AB	CES	AB	CES
0 (baseline)	0.0%	123	95	0	0	0.0	0.0
0.001	10.6%	110	80	-13	-15	+1.3	+3.5
0.005	53.2%	95	73	-28	-22	+3.6	+6.3
0.010	106.4%	94	73	-29	-22	+4.2	+6.3

<i>Panel B. EU-market exits by firm under AB pricing</i>					
$\Delta\phi$	AWS	MSFT	GOOG	ORCL	BABA
0.001	-1	-3	-3	-6	-1
0.005	-6	-10	-9	-6	-1
0.010	-6	-11	-9	-6	-1

Notes: The median calibrated market-level fixed cost in the preferred gravity baseline is 0.0094. The second column reports $\Delta\phi/0.0094$. Active, Δn , and profit changes are reported separately for AB and CES pricing, with profit changes measured relative to each pricing regime's own baseline. Profit changes are producer-profit changes. Panel B reports EU-market exits only.

Panel B shows that exits are concentrated in particular providers rather than spread evenly across the network. At the small wedge, Oracle loses the most EU deployments; at larger wedges, Microsoft and Google lose the largest numbers, while Alibaba has little EU presence to lose and AWS loses fewer EU deployments early on. The provider decomposition therefore reinforces the same margin: the compliance shock primarily

prunes firms with broader marginal EU exposure rather than uniformly shrinking all cloud networks. The difference between Panel B and aggregate Δn reflects net non-EU portfolio responses outside the EU.

We next consider a China access exercise, motivated by digital restrictions that can change feasible access rather than simply raise the cost of serving an already-feasible market. Restrictions such as the Great Firewall and app bans can affect which foreign services are available and which domestic substitutes users face (Sun, 2024; Sun and Trefler, 2023). In the model, this distinction maps into two primitives: a cost wedge changes profitability conditional on a market being feasible, while a feasible-set change alters the provider’s opportunity set itself.

Starting from the preferred gravity equilibrium, lowering a fixed-cost-like China wedge alone leaves total active pairs unchanged at 123. Expanding Alibaba’s feasible set to all 42 markets instead raises active pairs to 131 and gives Alibaba 13 additional markets. The exercise therefore treats digital fragmentation as a feasible-access margin rather than only a cost wedge.

The three exercises move different primitives but operate on the same threshold system. The AB-versus-CES counterfactual changes the pricing rule, the EU exercise changes regional deployment costs, and the China exercise changes feasible market access. Together, they show that market access in cloud infrastructure is determined jointly by deployment portfolios, pricing feedback, and the set of feasible locations.

4.6 Robustness and interpretation

The main AB-versus-CES result is stable across the robustness checks reported in the appendix. Expanding the candidate sets to allow all five providers to enter all 42 countries preserves the mechanism: endogenous markups still sustain a larger active set than the constant-markup counterfactual, and the aggregate profit loss under constant markups remains about 31%. Headquarters-distance robustness exercises also preserve the sign and order of magnitude of the mechanism: across the reported grid, fit remains between 197 and 199 of 210 cells, AB-active pairs remain between 124 and 128, and replacing AB markups with CES markups reduces active pairs by 19 to 29 and aggregate profit by about 30% to 32%.

The application should be interpreted through the equilibrium object developed in

the theory. In a combinatorial entry game with endogenous pricing, a small change in activation probabilities can move market shares, markups, and the profitability of marginal sites. The threshold-map argument is what makes the counterfactual well defined within the maintained strategy class. The gravity implementation then asks whether the same logic has quantitative bite once cloud regions are treated as access, latency, and residency assets.

The evidence is consistent across the main diagnostics and robustness checks. The activation matrix provides a sharp cross-sectional target, and the holdout exercise shows discipline within the same design. Richer identification would require provider-country revenue, country-level cloud spending, and region-entry timing. The solved-specification pricing and Jia–Tarski diagnostics show that the preferred gravity calibration lies in a numerically regular region of the maintained threshold system; the analytic uniqueness verification is provided by the benchmark section.

5. CONCLUSION

Firms that choose deployment portfolios do not solve independent binary-entry problems. Each activation changes the set of locations from which the firm can serve demand, the resulting portfolio changes market shares and markups, and those markups feed back into the profitability of other candidate locations. This paper studies when that feedback loop has a unique equilibrium and how much it matters quantitatively.

The theoretical contribution is a threshold-map approach to equilibrium selection. Within the maintained class of site-independent threshold strategies, equilibrium is a fixed point in activation probabilities. Uniqueness follows when this threshold map is a contraction, and the map’s Jacobian separates private-cost smoothing, marginal-value sensitivity, and threshold location. We verify the required objects in a Fréchet–CES–Atkeson–Burstein benchmark. There, the pricing block is not only a source of strategic interaction: when the Perron–Frobenius pricing-feedback object is inside the contraction region, pricing has a unique smooth selection and contributes a bounded amplification term to the extensive-margin Jacobian.

The quantitative contribution is to show that this pricing feedback matters for cloud infrastructure deployment. In the preferred gravity calibration, five hyperscale providers

choose which of 42 countries to equip with cloud regions. Holding fixed the calibrated accessibility and fixed-cost blocks, replacing endogenous Atkeson–Burstein markups with the constant CES markup reduces active provider-country pairs from 123 to 95 and lowers aggregate profit by 33.8%. Endogenous pricing therefore sustains 28 additional active provider-country pairs in the calibrated cloud economy. The EU compliance and China access exercises point to the same deployment margin: regional frictions can move near-threshold cells quickly, and feasible-market access can matter as much as cost relief.

The theorem establishes uniqueness within the maintained threshold-strategy class, and the cloud application quantifies that mechanism in a calibrated environment. Natural extensions include unrestricted Bayesian Nash equilibria, dynamic deployment in the spirit of industry-dynamics models such as Ericson and Pakes (1995); Weintraub et al. (2008); Igami and Yang (2016), where region entry is sequential and current portfolios shape future option values, as well as richer price and revenue data and a welfare analysis that combines market-access effects with consumer surplus, compliance benefits, and data-residency objectives.

In portfolio-entry environments, pricing feedback is not a detail to be appended after the entry problem is solved. It is part of the equilibrium object. Modeling that feedback changes both the uniqueness question and the counterfactual prediction for market access.

APPENDIX

Appendix organization. Appendix A contains the full proofs for the Fréchet–CES–Atkeson–Burstein benchmark verification. Appendix B contains the compact cloud application supplement on sample construction, candidate sets, and preferred calibration. The online appendix contains the remaining supplementary verification, diagnostics, robustness exercises, regional and access counterfactuals, algorithm diagnostics, and synthetic stress tests. In particular, Online Appendix [OA.2](#) records the nested-logit Bertrand portability check; Online Appendix [OA.3](#) records the gravity-access diagnostic objects used in the cloud application; and Online Appendix [OA.4](#) reports extended validation and comparator evidence.

A. PROOFS FOR THE CES–AB BENCHMARK

This appendix proves the benchmark verification results used in Section 3. The proofs follow the same logic as the main text. First, the pricing block is a Perron–Frobenius contraction in a weighted sup norm. Second, that pricing contraction gives a smooth pricing selection and bounds the pricing contribution to the marginal-value Jacobian. Third, the small-noise tail is localized around the complete-information equilibrium.

Throughout this appendix, σ denotes the within-sector CES elasticity and σ_ε denotes the private fixed-cost noise scale.

A.1 Proof of Proposition 8

Fix an activation profile. Let D collect the rival-demand indices $D_{-f}(n)$ across active firm-market pairs, and let $\Psi(D)$ denote the one-step AB pricing update in those indices. Because the active set is finite, marginal costs are strictly positive for active firms, and AB markups lie in the compact interval

$$\left[\frac{\sigma}{\sigma - 1}, \frac{\eta}{\eta - 1} \right],$$

the relevant pricing state space can be taken to be a compact positive box $\mathcal{D}$.

The proof uses the domination condition

$$|J_\Psi(D)| \leq \bar{\Lambda} \quad \text{for all } D \in \mathcal{D},$$

where $\bar{\Lambda}$ is nonnegative and $\rho(\bar{\Lambda}) < 1$. Fix any $\lambda_\Lambda \in (\rho(\bar{\Lambda}), 1)$. By the Perron–Frobenius Collatz–Wielandt argument, there exists a strictly positive vector $h \gg 0$ such that

$$\bar{\Lambda}h \leq \lambda_\Lambda h.$$

Define the weighted sup norm

$$\|x\|_h \equiv \max_j \frac{|x_j|}{h_j}.$$

For any x ,

$$|J_\Psi(D)x| \leq \bar{\Lambda}|x| \leq \bar{\Lambda}h \|x\|_h \leq \lambda_\Lambda h \|x\|_h.$$

Hence

$$\|J_\Psi(D)x\|_h \leq \lambda_\Lambda \|x\|_h \quad \text{for all } D \in \mathcal{D}.$$

By the mean-value theorem,

$$\|\Psi(D) - \Psi(D')\|_h \leq \lambda_\Lambda \|D - D'\|_h$$

for all $D, D' \in \mathcal{D}$. Thus Ψ is a contraction in the weighted sup norm. Banach's fixed-point theorem gives a unique pricing fixed point.

The same contraction margin gives smoothness. Let

$$F(D, c) = D - \Psi(D, c),$$

where c is the vector of active firms' marginal costs. At the fixed point, $F(D^*(c), c) = 0$. The derivative with respect to D is

$$D_D F(D^*(c), c) = I - J_\Psi(D^*(c), c).$$

Because $\|J_\Psi\|_h \leq \lambda_\Lambda < 1$, the inverse exists and satisfies

$$\|(I - J_\Psi)^{-1}\|_h \leq \frac{1}{1 - \lambda_\Lambda}.$$

The implicit-function theorem therefore gives a continuously differentiable pricing selection $D^*(c)$, and hence a continuously differentiable equilibrium price selection $p^*(c)$, on the maintained interior region.

It remains to verify the stated direct sup-norm sufficient condition. Under the normalization in (3.3), the off-diagonal weights in each active market row sum to one:

$$\sum_{g \neq f} \frac{p_{ng}(D)^{1-\sigma}}{D_{-f}(n; D)} = 1.$$

Thus the row sum of the pricing interaction matrix is bounded by $\max_{n,f} |\kappa_{nf}(D)|$. Cross-

market entries and diagonal entries are zero. Therefore

$$\|J\Psi(D)\|_\infty \leq \max_{n,f} |\kappa_{nf}(D)|.$$

If $\sup_D \max_{n,f} |\kappa_{nf}(D)| < 1$, then Ψ is also a direct sup-norm contraction. This proves Proposition 8. $\square$

A.2 Smoothness of the benchmark marginal-value map

We next prove the smoothness part of Proposition 10. Fix a firm-site pair (i, f) . Let $\mathcal{Z}_{if}$ denote the finite set of activation profiles for all firm-site pairs other than (i, f) . For an element $z \in \mathcal{Z}_{if}$, let $\mathcal{I}_f^{-i}(z)$ be firm f 's activation set excluding site i , and let $\mathcal{I}_{-f}(z)$ be the rivals' activation profile.

The deterministic profit increment from activating site i , net of the candidate site's private shock, is

$$\Delta\Pi_{if}^0(z) = \sum_n \left[VP_{nf}(\mathcal{I}_f^{-i}(z) \cup \{i\}, \mathcal{I}_{-f}(z)) - VP_{nf}(\mathcal{I}_f^{-i}(z), \mathcal{I}_{-f}(z)) \right] - \phi_i w_i.$$

For each realized activation profile z , the pricing equilibrium is unique and smooth by Proposition 8. Therefore $\Delta\Pi_{if}^0(z)$ is a well-defined finite number for every z .

Under the site-independent threshold restriction, the probability of profile z is a product of Bernoulli probabilities:

$$P_v(z) = \prod_{\ell: z_\ell=1} v_\ell \prod_{\ell: z_\ell=0} (1 - v_\ell).$$

The benchmark marginal-value index is

$$v_{if}(v) = \frac{1}{w_i} \sum_{z \in \mathcal{Z}_{if}} \Delta\Pi_{if}^0(z) P_v(z).$$

This expression is a finite multilinear polynomial in v . It therefore extends to a continuously differentiable function on an open neighborhood of $[0, 1]^M$. Since there are finitely many firm-site pairs, the full map $v(v)$ is C^1 on an open neighborhood of $[0, 1]^M$.

A.3 Bound on the marginal-value Jacobian

We now bound J_v . The derivative of v_{if} with respect to an activation probability v_ℓ can be written as a discrete difference. Let $z_{-\ell}$ denote all coordinates of the activation profile other than ℓ . Then

$$\frac{\partial v_{if}(v)}{\partial v_\ell} = \frac{1}{w_i} \mathbb{E}_{v_{-\ell}} \left[\Delta \Pi_{if}^0(z_\ell = 1, z_{-\ell}) - \Delta \Pi_{if}^0(z_\ell = 0, z_{-\ell}) \right].$$

Thus the Jacobian is bounded once the effect of toggling any other firm-site pair on the deterministic profit increment is uniformly bounded.

This effect has two components. The first is a direct cost and market-access component: adding a site changes Fréchet cost aggregates and variable profits holding the pricing response fixed. Because the set of firms, sites, and markets is finite, input costs and market sizes are bounded, and markups are bounded, this direct component is bounded by a finite constant L_C .

The second component is the pricing channel. Toggling an activation coordinate changes the cost vector entering the AB pricing block. Let Δc denote the resulting cost perturbation in the relevant active profile. The pricing fixed point satisfies

$$D^*(c) = \Psi(D^*(c), c).$$

Differentiating and using the inverse bound from Appendix A.1 gives

$$\|D^*(c + \Delta c) - D^*(c)\| \leq \frac{L_D}{1 - \lambda_\Lambda} \|\Delta c\|$$

for a finite primitive constant L_D , where the norm can be taken as any equivalent norm on the finite-dimensional pricing state space after adjusting the constant. Variable profits are Lipschitz in the induced pricing state on the maintained compact region; let L_μ denote a corresponding bound that includes the markup and share derivatives. Hence the pricing contribution to the marginal-value derivative is bounded by

$$\frac{L_D L_\mu}{1 - \lambda_\Lambda}.$$

Combining the direct and pricing components gives

$$\sup_{v \in [0,1]^M} \|J_v(v)\|_\infty \leq \frac{L_D L_\mu}{1 - \lambda_\Lambda} + L_C.$$

The constants L_D , L_μ , and L_C depend only on model primitives, markup bounds, market-size bounds, and the Fréchet cost aggregator. This proves the bounded-Jacobian part of Proposition 10.

A.4 Small-noise localization

We now prove the localized small-noise part. Let $x^{CI} \in \{0, 1\}^M$ be the unique complete-information equilibrium, and suppose the complete-information marginal-value gap is positive:

$$\delta_v = \min_m |v_m(x^{CI})| > 0.$$

By continuity of v , there exists a radius $r \in (0, 1/2)$ and a constant $b_0 > 0$ such that, on the cube

$$S = \{v : \|v - x^{CI}\|_\infty \leq r\},$$

we have

$$\min_m |v_m(v)| \geq b_0,$$

and the sign of $v_m(v)$ agrees with x_m^{CI} for every coordinate:

$$\mathbf{1}\{v_m(v) > 0\} = x_m^{CI}.$$

The threshold map is

$$T_{\sigma_\varepsilon, m}(v) = G_0\left(\frac{v_m(v)}{\sigma_\varepsilon}\right).$$

Because $|v_m(v)| \geq b_0$ uniformly on S , the CDF tails imply

$$\sup_{v \in S} |T_{\sigma_\varepsilon, m}(v) - x_m^{CI}| \rightarrow 0 \quad \text{as } \sigma_\varepsilon \rightarrow 0.$$

Therefore, for sufficiently small σ_ε ,

$$\|T_{\sigma_\varepsilon}(v) - x^{CI}\|_\infty \leq r \quad \text{for all } v \in S,$$

so $T_{\sigma_\varepsilon}(S) \subseteq S$. This proves invariance.

Next, suppose v is a fixed point outside S . By Assumption 9, there exists a coordinate m such that

$$|v_m(v)| \geq b, \quad |\mathbf{1}\{v_m(v) > 0\} - v_m| \geq 2r.$$

Since $|v_m(v)| \geq b$, for sufficiently small σ_ε ,

$$|T_{\sigma_\varepsilon, m}(v) - \mathbf{1}\{v_m(v) > 0\}| \leq r.$$

But if v is a fixed point, then $v_m = T_{\sigma_\varepsilon, m}(v)$, which implies

$$|\mathbf{1}\{v_m(v) > 0\} - v_m| \leq r,$$

contradicting the isolation condition. Thus every fixed point must lie in S for all sufficiently small σ_ε . This proves fixed-point inclusion.

Finally, uniform non-marginality on S follows directly from $\min_m |v_m(v)| \geq b_0$. Hence S is invariant, fixed-point-inclusive, and uniformly non-marginal for all sufficiently small σ_ε .

A.5 Proof of Proposition 10

We combine the pieces.

Part (i) follows from Appendix A.2. The marginal-value map is a finite Bernoulli mixture of deterministic profit increments. Each profit increment is well defined because the pricing block has a unique smooth selection by Proposition 8. The Bernoulli weights are multilinear in activation probabilities, so v is C^1 on an open neighborhood of $[0, 1]^M$.

Part (ii) follows from Appendix A.3. The derivative of each marginal value with respect to any activation probability is a finite expectation of a toggle effect. That toggle effect is bounded by a direct Fréchet-cost and payoff component, summarized by L_C , plus an indirect pricing component. The indirect component is bounded by the Perron–Frobenius

pricing response:

$$\frac{1}{1 - \lambda_\Lambda}.$$

Thus

$$K_v \equiv \sup_{v \in [0,1]^M} \|J_v(v)\|_\infty \leq \frac{L_D L_\mu}{1 - \lambda_\Lambda} + L_C.$$

Part (iii) follows from Appendix A.4. A unique complete-information equilibrium with positive marginal-value gap gives local sign stability. Assumption 9 rules out fixed points outside the local cube. Together they imply that, for all sufficiently small σ_ε , the localized set S is invariant, fixed-point-inclusive, and uniformly non-marginal.

These three parts verify the smoothness, bounded-Jacobian, and small-noise localization ingredients required by the abstract theorem. If the resulting threshold-map contraction modulus satisfies $\gamma(\sigma_\varepsilon) < 1$, then the threshold-strategy BNE is unique within the maintained site-independent threshold-strategy class. If

$$\gamma_{\max} = \sup_{\sigma_\varepsilon > 0} \gamma(\sigma_\varepsilon) < 1,$$

then uniqueness holds uniformly over private-cost noise scales. $\square$

A.6 Interpretation of the proof

The proof separates three objects that are often conflated. The first is pricing selection. Proposition 8 shows that the AB pricing block is unique and smooth when the Perron–Frobenius pricing-feedback object is inside the contraction region. The second is extensive-margin sensitivity. The same pricing contraction appears in the marginal-value Jacobian through the amplification term $1/(1 - \lambda_\Lambda)$. The third is localization. A positive complete-information gap gives local sign stability, but fixed-point inclusion requires the additional benchmark isolation condition.

This separation is useful for the cloud application. The benchmark provides the analytic proof of the theorem’s objects. Appendix B then records the compact cloud application supplement used directly in the main manuscript, while the online appendix records the remaining supplementary checks: nested-logit pricing portability, gravity-access diagnostics, extended validation and comparator evidence, regional and access counterfactuals, algorithm diagnostics, and synthetic stress tests. The main text therefore

keeps distinct the objects that are proved analytically, the compact application supplement, and the deeper diagnostics and counterfactuals used to support the calibrated specification.

B. CLOUD APPLICATION: SAMPLE AND CALIBRATION

This appendix records the compact application-side details used directly in the main manuscript. It documents how the provider-country panel is coded, how candidate sets are constructed, and how the preferred gravity calibration is disciplined before the counterfactuals in Section 4 are computed. The online appendix reports the remaining extended validation, robustness, and comparator exercises.

B.1 Sample construction and candidate sets

The empirical panel is provider-country. A cell is coded active when official provider documentation lists at least one operational cloud region in that country. Pipeline status is recorded separately when official documentation lists a future region. Datacenter Map is used as a cross-check and for harmonized country coding; when the two sources disagree, provider documentation takes precedence.

Candidate sets are constructed from operational plus pipeline country-cells. This is a maintained lower-bound design. It does not add hypothetical feasible countries to improve fit. It also means that cells outside a provider’s candidate set are mechanically inactive by construction rather than economically chosen inactive.

Table 6 decomposes the preferred-gravity fit. The full panel has 210 provider-country cells. Of these, 134 lie inside the maintained feasible sets and 76 lie outside. The preferred gravity baseline matches 200 of 210 cells in the full panel and 124 of 134 feasible cells.

Table 6: Feasible-Set Decomposition of the Provider-Country Panel

Sample	Cells	Correctly classified	Accuracy
Full panel	210	200	95.2%
Feasible cells only	134	124	92.5%
Operational cells	129	121	93.8%
Pipeline cells	5	3	60.0%
Outside feasible set	76	76	mechanically inactive

Notes: The feasible-set lower bound consists of 129 operational country-cells and 5 pipeline country-cells. The outside-feasible row is imposed by the maintained candidate-set design and is not a free prediction of the model.

The feasible-cell decomposition is important for interpretation. Full-panel fit is useful

for summarizing the activation matrix, but feasible-cell fit isolates the part of the classification exercise where the model is making economically informative activation choices.

B.2 Preferred gravity calibration

Table 7 records the accessibility inputs used in the preferred gravity baseline. These objects are fixed before calibrating the fixed-cost block. They are chosen on a small plausibility grid rather than estimated from the binary entry moments.

Table 7: Preferred Gravity Baseline: Accessibility Inputs

Object	Value	Role
β_d	0.25	distance penalty in the gravity score
β_{same}	0.20	same-country bonus
β_{local}	0.15	local-eligibility / residency bonus
β_{policy}	0.20	digital-policy loading
λ_A	1.50	log-sum smoothness in accessibility
χ	0.40	accessibility-to-quality loading in the preferred baseline
β	0.70	concavity of the cloud-proxy market-size index Ω_n
Normalization	difference	frontier-difference accessibility normalization
Policy mode	suggested_weight	OECD policy aggregation rule

Notes: The preferred baseline starts from the G3 gravity specification and raises χ to 0.40. These inputs are fixed outside the entry-moment calibration; the fixed-cost block is calibrated conditional on them.

Given these inputs, the fixed-cost block (ϕ_i, g_f) is calibrated by alternating block coordinate descent. The inner loop updates market-level deployment costs ϕ_i to match per-market entry counts,

$$m_i = \sum_f I_{fi},$$

while the outer loop updates $\gamma_f = \log g_f$ to match per-firm totals,

$$m_f = \sum_i I_{fi}.$$

The objective is the combined entry-count criterion

$$\sum_i \left(m_i^{\text{data}} - m_i^{\text{model}} \right)^2 + \sum_f \left(m_f^{\text{data}} - m_f^{\text{model}} \right)^2.$$

The implementation normalizes $g_{\text{AWS}} = 1$, regularizes ϕ_i toward its prior, and in the preferred gravity runs imposes bounds

$$\phi_i \in [0.005, 0.50], \quad g_f \in [0.10, 5.0].$$

This is the sense in which the application is calibrated from entry moments rather than separately identified from revenue, price, or region-entry timing data.

Table 8 reports the calibrated provider-level fixed-cost shifters and the per-firm effect of the main AB-versus-CES counterfactual. The preferred gravity baseline has 123 AB-active provider-country pairs. Replacing AB markups with the constant CES markup eliminates 28 of them.

Table 8: Preferred Gravity Baseline: Fixed Costs and AB–CES Incidence

Firm	g_f	AB active cells	CES-lost cells	CES profit change (%)
AWS	1.000	25	9	-30.5
MSFT	0.472	33	6	-31.7
GOOG	0.419	29	6	-43.6
ORCL	0.181	26	4	-46.2
BABA	0.244	10	3	-48.0

Notes: The table reports the preferred-gravity baseline values of the multiplicative fixed-cost wedge g_f , the number of AB-active country-cells, and the number of country-cells lost when AB markups are replaced by CES markups. The market-level fixed-cost distribution is: min = 0.0050, 25th percentile = 0.0050, median = 0.0094, 75th percentile = 0.0269, mean = 0.0208, max = 0.1200.

The main text reports the corresponding fit, holdout, and solved-specification diagnostics. Online Appendix [OA.4](#) reports the remaining extended validation exercises and comparator evidence.

REFERENCES

- Antràs, Pol, Teresa C. Fort, and Felix Tintelnot, “The Margins of Global Sourcing: Theory and Evidence from US firms,” *American Economic Review*, 2017, 107 (9), 2514–2564.
- , —, Evgenii Fadeev, and Felix Tintelnot, “Exporting, Global Sourcing, and Multinational Activity: Theory and Evidence from the United States,” *Review of Economics and Statistics*, 2026.

- Arkolakis, Costas, Fabian Eckert, and Rowan Shi**, “Combinatorial Discrete Choice: Theory and Application to Multinational Production,” *NBER Working Paper 31877*, 2023.
- Athey, Susan**, “Single Crossing Properties and the Existence of Pure Strategy Equilibria in Games of Incomplete Information,” *Econometrica*, 2001, 69 (4), 861–889.
- Atkeson, Andrew and Ariel Burstein**, “Pricing-to-Market, Trade Costs, and International Relative Prices,” *American Economic Review*, 2008, 98 (5), 1998–2031.
- Berry, Steven T.**, “Estimation of a Model of Entry in the Airline Industry,” *Econometrica*, 1992, 60 (4), 889–917.
- Bresnahan, Timothy F. and Peter C. Reiss**, “Entry and Competition in Concentrated Markets,” *Journal of Political Economy*, 1991, 99 (5), 977–1009.
- Castro-Vincenzi, Juanma**, “Climate Hazards and Resilience in the Global Car Industry,” *Working Paper*, 2024.
- Ciliberto, Federico and Elie Tamer**, “Market Structure and Multiple Equilibria in Airline Markets,” *Econometrica*, 2009, 77 (6), 1791–1828.
- Demirer, Mert, Diego Jiménez-Hernández, Dean Li, and Sida Peng**, “Data, Privacy Laws and Firm Production: Evidence from the GDPR,” 2024. NBER Working Paper No. 32146.
- Doraszelski, Ulrich and Mark Satterthwaite**, “Computable Markov-Perfect Industry Dynamics,” *RAND Journal of Economics*, 2010, 41 (2), 215–243.
- Edmond, Chris, Virgiliu Midrigan, and Daniel Yi Xu**, “Competition, Markups, and the Gains from International Trade,” *American Economic Review*, 2015, 105 (10), 3183–3221.
- Ericson, Richard and Ariel Pakes**, “Markov-Perfect Industry Dynamics: A Framework for Empirical Work,” *Review of Economic Studies*, 1995, 62 (1), 53–82.
- Espin-Sanchez, Jose-Antonio, Alvaro Parra, and Yuzhou Wang**, “Equilibrium Uniqueness in Entry Games with Private Information,” *RAND Journal of Economics*, 2023, 54 (3), 512–540.

- Fan, Ying and Chenyu Yang**, “Competition, Product Proliferation, and Welfare: A Study of the U.S. Smartphone Market,” *American Economic Journal: Microeconomics*, 2020, 12 (2), 99–134.
- Fang, Tommy Pan and Shane Greenstein**, “Where the Cloud Rests: The Economic Geography of Data Centers,” *Strategy Science*, 2025, 10 (4), 404–420.
- Gans, Joshua S., Mikael Hervé, and Muath Masri**, “Economic Analysis of Proposed Regulations of Cloud Services in Europe,” *European Competition Journal*, 2023, 19 (3), 522–568.
- Holmes, Thomas J.**, “The Diffusion of Wal-Mart and Economies of Density,” *Econometrica*, 2011, 79 (1), 253–302.
- Igami, Mitsuru and Nathan Yang**, “Unobserved Heterogeneity in Dynamic Games: Cannibalization and Preemptive Entry of Hamburger Chains in Canada,” *Quantitative Economics*, 2016, 7 (2), 483–521.
- Jia, Panle**, “What Happens When Wal-Mart Comes to Town: An Empirical Analysis of the Discount Retailing Industry,” *Econometrica*, 2008, 76 (6), 1263–1316.
- Manganelli, Antonio and Daniel Schnurr**, “Competition and Regulation of Cloud Computing Services: Economic Analysis and Review of EU Policies,” 2024. Working paper.
- Milgrom, Paul and Chris Shannon**, “Monotone Comparative Statics,” *Econometrica*, 1994, 62 (1), 157–180.
- Nishida, Mitsukuni**, “Estimating a Model of Strategic Network Choice: The Convenience-Store Industry in Okinawa,” *Marketing Science*, 2015, 34 (1), 20–38.
- Nocke, Volker and Nicolas Schutz**, “Multiproduct-Firm Oligopoly: An Aggregative Games Approach,” *Econometrica*, 2018, 86 (2), 523–557.
- Seim, Katja**, “An Empirical Model of Firm Entry with Endogenous Product-Type Choices,” *The RAND Journal of Economics*, 2006, 37 (3), 619–640.
- Sun, Ruiqi**, “Blocking the Giants: Theory and Evidence from App Bans,” 2024. Working paper.

— and **Daniel Trefler**, “The Impact of AI and Cross-Border Data Regulation on International Trade in Digital Services: A Large Language Model,” 2023. NBER Working Paper No. 31925.

Tintelnot, Felix, “Global Production with Export Platforms,” *Quarterly Journal of Economics*, 2017, 132 (1), 157–209.

Weintraub, Gabriel Y., C. Lanier Benkard, and Benjamin Van Roy, “Markov Perfect Industry Dynamics with Many Firms,” *Econometrica*, 2008, 76 (6), 1375–1411.

Wollmann, Thomas G., “Trucks without Bailouts: Equilibrium Product Characteristics for Commercial Vehicles,” *American Economic Review*, 2018, 108 (6), 1364–1406.